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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

RI2111-1 – SMART MOBILE ROBOTS
(For RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Degree of maneuverability is _____
A) equal to degree of freedom
B) equal to degree of steerability
C) equal to degree of mobility and degree of freedom
D) equal to sum of degree of mobility and degree of steerability
2. The vector of the input velocity command of the land-based mobile robot is given as $\zeta = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and the mobile robot (body) frame is oriented 90° with respect to the Fixed frame. In this case, what will be the time derivatives of the generalized coordinates?
A) $\dot{\eta} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$
B) $\dot{\eta} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$
C) $\dot{\eta} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$
D) $\dot{\eta} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$
3. The concept of "inverse differential kinematics" in mobile robots is used to:
A) Calculate the wheel rotations required to achieve a desired robot position
B) Determine the robot's speed based on its wheel rotation
C) Measure the degree of freedom of the robot
D) Calculate the energy consumption based on the robot's movement
4. For a four wheel mecanum wheel drive mobile robot: the degree of mobility is, ____, and the degree of steerability is, ____
A) 4 and 4
B) 0 and 4
C) 4 and 0
D) 4 and 3
5. In a two fixed wheel (differential) drive mobile robot, the left side wheel angular velocity is 0.5 rad/s and the right side wheel angular velocity is 1.5 rad/s, both wheels are same size and the wheel radius is 10 cm. What will be the longitudinal velocity (u) of the robot with respect to the body frame?
A) 0.2 m/s
B) 0.1 m/s
C) 2 m/s
D) 0.05 m/s
6. If the controllable degrees of freedom are equal to total degrees of freedom, then the system is called
A) Mobile robot
B) Holonomic system
C) Nonholonomic system
D) Differential drive

7. For a given desired trajectory, η , $\dot{\eta}$, $\ddot{\eta}$, finding the required input vector τ is known as
- A) Forward kinematics
B) Forward dynamics
C) Inverse kinematics
D) Inverse dynamics
8. A Mobile Robot is considered 'Holonomic' if:
- A) It uses a tricycle wheel configuration
B) It has a fixed number of wheels
C) It can only move in the direction of one of its axes at a time
D) It can move in any direction in its workspace without rotating
9. Tactile sensors allow a robot to:
- A) Hear sounds in the environment
B) Feel and respond to physical contact
C) Smell chemical substances
D) See colors and patterns
10. What is the primary function of sensors in mobile robots?
- A) To provide energy
B) To increase speed
C) To perceive the environment
D) To reduce cost
11. What does the Lagrange-Euler Formulation in mobile robot dynamics primarily describe?
- A) To program the AI of robots
B) The cost of robot production
C) The relation between force and its motion
D) The color and design of the robot
12. What does the Kalman filter algorithm primarily deal with?
- A) Maximization of system throughput in communication channel
B) Estimation of the state of a linear system from noisy measurements
C) Encryption and decryption of data in secure communication
D) Optimization of non-linear functions
13. What does LiDAR stand for?
- A) Linear Detection and Ranging
B) Light Direction and Ranging
C) Light Detection and Ranging
D) Linear Distance and Ranging
14. Choose the false statement:
- A) Prediction update is based on the updated belief state
B) Perception update decreases the uncertainty as the robot moves
C) Action update is based on the prior belief state.
D) Prediction update increases uncertainty as the robot moves
15. In the context of Markov localization, what is a 'belief'?
- A) The probability distribution over all possible locations of the robot
B) A fixed point on the map where the robot is programmed to go
C) The robot's confidence in its abilities.
D) The robot's preference for certain paths.
16. Lagrangian (L) is equal to _____
- A) Kinetic energy + Potential energy
B) Kinetic energy - Potential energy
C) Kinetic energy $\times$ Potential energy
D) Kinetic energy / Potential energy
17. What is the primary purpose of a visibility graph in robot path planning?
- A) To find the shortest path that avoids obstacles
B) To create a dense network of nodes and edges
C) To construct a graph that ignores obstacles
D) To navigate without using any sensors
18. The Voronoi diagram in path planning is used to:
- A) Minimize the distance to the goal
B) Ensure optimal speed
C) Simplify the robot's shape for planning
D) Maximize the distance from obstacles
19. In the context of path planning, what does "completeness" refer to
- A) The total computational power required for planning
B) The robot system's ability to reach a goal state when possible
C) The ability to avoid all obstacles
D) The accuracy of the robot's sensors
20. A repulsive potential field is designed to
- A) Attract the robot to the goal
B) Slow down the robot near obstacles
C) Move the robot away from obstacles
D) Simplify obstacle shapes

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define forward kinematics and inverse kinematics in the context of mobile robots. Derive an expression of the forward kinematics ($\dot{\eta} = J(\psi)\zeta$) for a mobile robot? b) Consider a mobile robot equipped with four Mecanum wheels and initially oriented parallel to the x-axis. The robot is required to move clockwise direction about its centre, determine the appropriate combination of wheel angular velocities? Provide a proof to support your answer.	8	L*2	1	1
2.	a) Compare and contrast wheeled, legged, tracked, and hybrid locomotion mechanisms used in mobile robots. Discuss their advantages and limitations. b) What are the characteristics of an omni-wheel mobile robot? For a given mobile robot with three omni-directional wheels, if angular velocities are $\begin{bmatrix} 0.5 \\ -0.25 \\ -0.25 \end{bmatrix}$ rad/s, $a=0.2m, l = 0.3m$ find the longitudinal velocity (u) lateral velocity (v) and angular velocity (r) of the robot with respect to the body frame?	8	L2	2	2
3.	a) What is the degree of mobility, steerability and maneuverability of a mobile robot? Find the degree of Mobility, degree of Steerability and maneuverability for the following wheeled mobile robot configurations. The solid shaded wheels are powered and hatched lines are not powered wheels as shown in Fig Q3(a). (i) (ii) (iii) Fig. Q3(a)	8	L3	2	2
	b) What is a skid-steering wheel mobile robot? In a skid-steering mobile robot, the robot moves with a linear velocity $u=0.6$ m/s and an angular velocity $r=0.3$ rad/s with respect to the body frame. The robot has four wheels of the same size, each with a radius of 10 cm. The wheels are placed symmetrically along the y-axis of the body frame, with a track width (distance between the left and right wheels) of 50 cm. The robot follows a differential steering principle by controlling the speed of the left and right wheel pairs independently. Determine the angular velocities of the left and right wheel pairs, respectively.	8	L3	1	2
Unit – II					
4.	a) Derive dynamic equation of motion for a differential wheeled mobile robot with two independently driven wheels. b) What is the Kalman Filter localization, and how does it assist in mobile robot localization? Explain	8	L3	3	1
		8	L2	4	1

5. a) How are sensors used for wheeled mobile robots classified? Explain and Give examples for each type
- b) How does dead reckoning differ from odometry, and what are their respective applications in robot localization?
6. a) Derive dynamic equation of motion for a mecanum wheel mobile robot.
- b) Discuss the challenges of localization in mobile robots. How do sensor noise, sensor aliasing, and effector noise each contribute to localization errors, and what strategies can be employed to mitigate these issues?

8	L2	3	1
8	L2	4	1
8	L3	3	1
8	L2	4	1

Unit – III

7. a) Explain the principles of the bug algorithm and its application in enabling mobile robots to navigate around obstacles
- b) How does the Voronoi diagram method work for path planning, and how is it used to generate paths that maximize distance from obstacles?
8. a) Explain cell decomposition techniques used in path planning.
- b) Given the graph representing a map for a mobile robot in Fig.Q8(b), where nodes correspond to waypoints, and edges represent possible paths with their associated travel costs, employ the A* search algorithm to find the most cost-efficient route for the robot to travel from the start node S to the goal node G.

8	L2	5	1
8	L2	5	1
8	L2	5	1

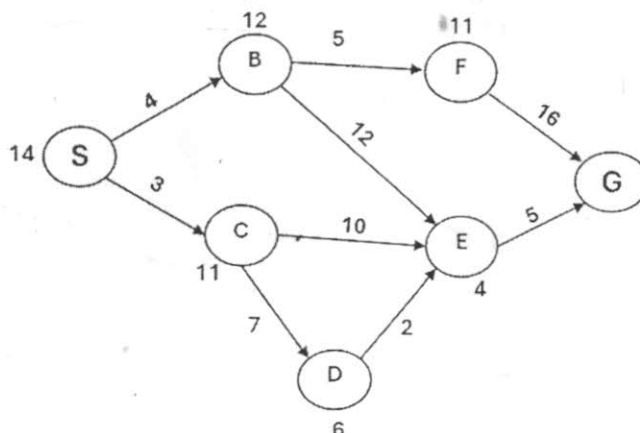


Fig. Q 8(b)

8	L3	5	2
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BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
